

GHANA MARITIME DEFENSE COMMAND (GMDC)

THREAT MODEL DOCUMENT

Threat Modeling Objective

This threat model identifies potential threats to GMDC's information systems and considers their possible effect on confidentiality, integrity, and availability. It covers users, endpoints, network infrastructure, public-facing services, applications, servers, and databases.

Threat Actors and Objectives

Threat Actor	Objective	Likely Target
External attacker	Unauthorized access, information theft or service disruption.	Web/email services, firewall, exposed services
Phishing/socialengineering attacker	Obtain credentials or cause unsafe user actions.	Employee PCs and email accounts
Malicious insider	Access, alter or disclose information beyond authorization.	Databases, files, applications
Compromised account	Use legitimate credentials to reach protected resources.	Internal applications and sensitive records
Malware/automated threat	Disrupt systems, steal information or gain unauthorized access.	Endpoints, file server and business systems

STRIDE Threat Analysis

Category	Threat Scenario	Impact	Recommended Control
Spoofing	Stolen credentials are used to impersonate an authorized employee.	Unauthorized access.	MFA, strong authentication, account monitoring.
Tampering	Unauthorized changes are made to data, files or configurations.	Loss of integrity; incorrect records.	RBAC, change control, audit logs, integrity monitoring.

Repudiation	Actions cannot be reliably traced to a responsible account.	Weak accountability and investigation.	Unique accounts, centralized logs, timestamps.
Information Disclosure	Sensitive personnel, financial or operational information is exposed.	Confidentiality breach.	Encryption, least privilege, access control, data classification.
Denial of Service	Public or internal services become unavailable because of malicious/excessive traffic.	Service disruption.	Firewall controls, rate limiting, monitoring, redundancy and recovery.
Elevation of Privilege	A low-privileged account obtains administrative permissions.	Unauthorized system/data control.	Least privilege, privileged-account separation, MFA, patching and monitoring.

Attack Surfaces / Entry Points

- Internet-facing web and email services located in the DMZ.
- Employee PCs and user credentials, including phishing and malicious attachments.
- Administrative access to network devices and servers.
- Internal applications and databases accessed by authorized users.
- File-sharing and backup infrastructure.
- Misconfigured or insufficiently protected network devices and security controls.

Risk Priorities and Security Measures

The highest priorities are privileged accounts, sensitive databases, critical applications, network infrastructure, and backups. Recommended controls include multi-factor authentication, role-based access control, network segmentation, secure configuration, regular patching, endpoint protection, centralized logging, protected backups, security monitoring, and user security awareness. Public-facing services should remain separated from critical internal systems through the DMZ and firewall.